

G - 221 Substring

分值: 600 分

Problem Statement

For a sequence $X = (X_1, \dots, X_n)$ of positive integers, we call X a **221 sequence** if and only if the length and value are equal for every (length, value) pair in its run-length encoding. More formally, a sequence satisfying the following condition is called a 221 sequence.

对于由正整数构成的序列 $X = (X_1, \dots, X_n)$, 当且仅当其游程编码中每一组 (长度, 数值) 的长度和数值相等时, 我们称 X 为**221序列**。更正式地, 满足以下条件的序列称为221序列。

- For any integer pair (l, r) satisfying $1 \leq l \leq r \leq n$, if all three of the following conditions hold, then $(r - l + 1) = X_l$.
 - $l = 1$ or $(l \geq 2 \text{ and } X_{l-1} \neq X_l)$
 - $r = n$ or $(r \leq n - 1 \text{ and } X_{r+1} \neq X_r)$
 - $X_l = X_{l+1} = \dots = X_r$
- 对于任意满足 $1 \leq l \leq r \leq n$ 的整数对 (l, r) , 若以下三个条件同时成立, 则 $(r - l + 1) = X_l$.
 - $l = 1$ 或 $(l \geq 2 \text{ 且 } X_{l-1} \neq X_l)$
 - $r = n$ 或 $(r \leq n - 1 \text{ 且 } X_{r+1} \neq X_r)$
 - $X_l = X_{l+1} = \dots = X_r$

For example, $(2, 2, 3, 3, 3, 1, 2, 2)$ is a 221 sequence, but $(1, 1)$ and $(4, 4, 1, 4, 4)$ are not 221 sequences.

例如, $(2, 2, 3, 3, 3, 1, 2, 2)$ 是221序列, 但 $(1, 1)$ 和 $(4, 4, 1, 4, 4)$ 不是221序列。

You are given a sequence $A = (A_1, \dots, A_N)$ of positive integers of length N . Find the number of distinct 221 sequences that appear as a non-empty **contiguous subsequence** of A .

给定一个长度为 N 的正整数序列 $A = (A_1, \dots, A_N)$, 求其中作为非空**连续子序列**出现的不同221序列的个数。

Even if two subsequences are extracted from different positions, they are counted as one if they are equal as sequences.

即使两个子序列从不同位置提取, 只要作为序列相等, 就只计为一次。

Constraints

- $1 \leq N \leq 500\,000$
- $1 \leq A_i \leq 9$
- All input values are integers.
- $1 \leq N \leq 500\,000$
- $1 \leq A_i \leq 9$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按以下格式从标准输入给出：

N
 $A_1 A_2 \cdots A_N$

Output

Output the answer on a single line.

将答案输出在一行中。

Sample Input 1

```
23
2 2 3 3 3 1 1 1 3 3 3 1 2 2 2 1 9 1 4 4 4 4 4
```

Sample Output 1

```
14
```

The 221 sequences that appear as a non-empty contiguous subsequence of A are the following 14:

作为A的非空连续子序列出现的221序列共有以下14个：

- (1)
- (1, 2, 2)
- (1, 3, 3, 3)
- (1, 3, 3, 3, 1)
- (1, 3, 3, 3, 1, 2, 2)
- (1, 4, 4, 4, 4)
- (2, 2)
- (2, 2, 1)
- (2, 2, 3, 3, 3)
- (2, 2, 3, 3, 3, 1)
- (3, 3, 3)
- (3, 3, 3, 1)
- (3, 3, 3, 1, 2, 2)
- (4, 4, 4, 4)
- (1)
- (1, 2, 2)
- (1, 3, 3, 3)
- (1, 3, 3, 3, 1)
- (1, 3, 3, 3, 1, 2, 2)
- (1, 4, 4, 4, 4)
- (2, 2)
- (2, 2, 1)
- (2, 2, 3, 3, 3)
- (2, 2, 3, 3, 3, 1)
- (3, 3, 3)

- (3, 3, 3, 1)
- (3, 3, 3, 1, 2, 2)
- (4, 4, 4, 4)

Sample Input 2

```
2
6 7
```

Sample Output 2

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0
```

No 221 sequences appear as a non-empty contiguous subsequence of A .

A 中不存在作为非空连续子序列出现的221序列。